

PROJECT DOCUMENTATION

Project title : EduGenie-AI

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1. INTRODUCTION

EduGenie is a lightweight AI-powered educational assistant designed to simplify learning through generative AI. It provides students with an interactive way to ask questions, understand difficult concepts, generate quizzes, summarize educational content and obtain personalized learning recommendations.

2. PROJECT DESCRIPTION

EduGenie is designed for students of different academic levels. It combines cloud-based AI capabilities with a simple web interface. The backend is developed using FastAPI and the frontend uses HTML and CSS. The system supports both lightweight local processing and cloud-based AI services.

- Ask questions and receive smart, concise answers.
- Understand complex concepts through simplified explanations.
- Generate quizzes from topics or text.
- Receive personalized learning recommendations.
- Summarize large educational passages.

3. OBJECTIVES

- Provide an accessible AI-based learning companion.
- Make complex educational concepts easier to understand.
- Support quick revision through summarization.
- Help students assess their understanding through quizzes.
- Provide structured learning paths from beginner to advanced levels.

4. SYSTEM REQUIREMENTS

4.1 Software Requirements

- Python 3.10+
- FastAPI Framework
- HTML and CSS
- Google Gemini API
- Uvicorn ASGI Server
- Jinja2 HTML Templating Engine

4.2 API Setup

The project documentation specifies creating a Google AI Studio project, enabling the Gemini API, generating an API key and storing it securely.

5. ARCHITECTURE AND TECHNOLOGY STACK

EduGenie uses FastAPI as the backend framework and a simple HTML/CSS frontend. Gemini 1.5 Pro is used through an API for question answering, summarization, quiz generation and learning paths. LaMini-Flan-T5-783M is used locally for concept explanation.

- FastAPI – backend REST API.
- HTML/CSS – web interface and styling.
- Gemini 1.5 Pro – cloud-based generative AI functions.
- LaMini-Flan-T5-783M – local concept explanation.
- Uvicorn – server for running the FastAPI application.
- Jinja2 – HTML templating.

6. PROJECT WORKFLOW

Milestone 1: Model Selection and Architecture

Gemini 1.5 Pro is selected for Q&A, summarization, quiz generation and learning paths. LaMini-Flan-T5-783M is selected for concept explanation because it is lightweight and CPU-compatible.

Milestone 2: Core Functionalities Development

The project implements separate modules for explanation, Q&A, quiz generation, summarization and learning recommendations. Each module is connected to a corresponding FastAPI endpoint.

Milestone 3: Frontend Development

The web interface contains a task dropdown, text area, submit button and result container. Form submission sends a POST request to the FastAPI backend and displays the result in real time.

Milestone 4: Deployment

The application can be run locally using Uvicorn with the command: **uvicorn main:app --reload**. It can then be accessed locally at **http://127.0.0.1:8000**.

7. FOLDER ARCHITECTURE

main.py – FastAPI application
explanation_module.py – concept explanation logic
qna.py – question answering
quiz_module.py – quiz generation
summary_module.py – summarization
learning_path.py – learning recommendations
templates/index.html – HTML frontend
static/style.css – styling
requirements.txt – Python dependencies

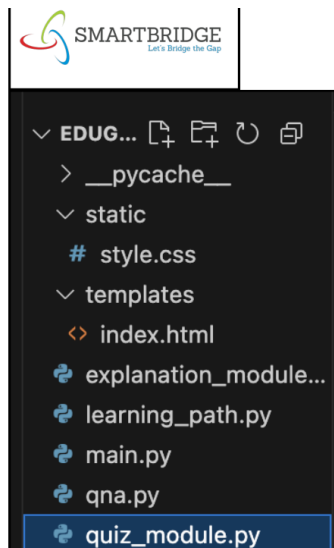


Fig. Folder Architecture – EduGenie project files and folders

8. MODULE DESCRIPTION

8.1 Explanation Module

The Explanation Module uses LaMini-Flan-T5 to provide concise, context-aware explanations. It breaks complex topics into easily understandable language and is particularly useful for beginners, school students and self-learners.

8.2 Q&A; Module

The Q&A; Module uses Gemini 1.5 Pro to answer general knowledge and academic questions across different subjects.

8.3 Quiz Module

The Quiz Module generates three multiple-choice questions from a given passage. Each question contains four options and a correct answer. The output is structured in JSON format for easy integration.

8.4 Summary Module

The Summary Module uses Gemini's generative capabilities to convert long paragraphs into concise and easy-to-understand versions while retaining the core information.

8.5 Learning Path Module

The Learning Path Module generates a structured learning path for a selected topic. It organizes concepts from beginner to advanced levels and can provide resources such as videos, articles or books.

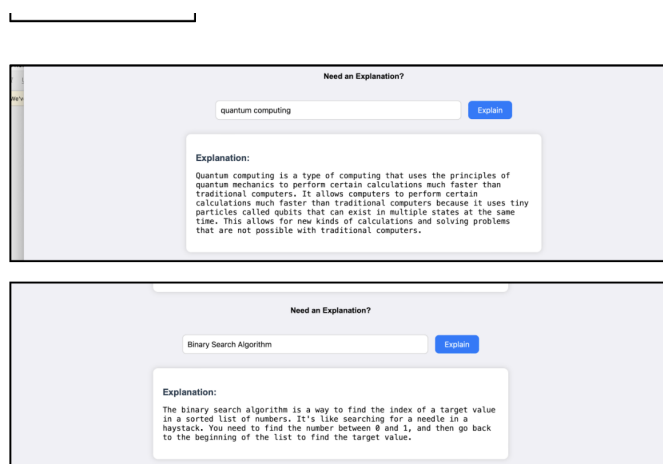


Fig. Explanation Module – actual project output

9. BACKEND API

The project defines RESTful endpoints for the major EduGenie functions:

- /qa
- /explain
- /quiz
- /summarize
- /learn/recommendations

Each endpoint is connected to its respective module logic.

- /learn/recommendations
- Connected each endpoint to a respective module logic

```
main.py
40 # QA - GET API using Gemini
41 @app.get("/qa")
42 async def answer_question(question: str = Query(...)):
43     answer = answer_question_with_gemini(question)
44     return {"answer": answer}
45
46 # Explaination - POST API
47 @app.post("/explain")
48 async def explain_api(request: Request):
49     data = await request.json()
50     topic = data.get("topic")
51     if not topic:
52         return JSONResponse(content={"error": "Please provide a topic."}, status_code=400)
53     explanation = explain_topic(topic)
54     return {"topic": topic, "explanation": explanation}
55
56 # Summarization - POST API
57 @app.post("/summarize")
58 async def summarize_api(request: Request):
59     data = await request.json()
60     text = data.get("text")
61     if not text:
62         return JSONResponse(content={"error": "Please provide text to summarize."}, status_code=400)
63     summary = summarize_text(text)
64     return {"summary": summary}
65
```

Fig. Endpoints of API

```

66 # Quiz Generation - POST API
67 @app.post("/quiz")
68 async def quiz_api(request: Request):
69     data = await request.json()
70     text = data.get("text")
71     if not text:
72         return JSONResponse(content={"error": "Please provide text for quiz."}, status_code=400)
73     quiz = generate_quiz(text)
74     print("Generated quiz:", quiz) # DEBUG
75     return JSONResponse(content={"quiz": quiz})
76
77 # Learning Recommendations - GET API
78 @app.get("/learn/recommendation")
79 async def learning_recommendation_api(topic: str = Query(...)):
80
81
```

Fig. Backend API / endpoint implementation from the project files

10. FRONTEND DEVELOPMENT

The frontend is built using HTML and CSS. The main interface provides a task selection dropdown with Explain, QnA, Quiz, Summary and Recommend Path options. Users enter their content in a text area and submit it for processing.

- Responsive design.
- Styled buttons and input areas.
- Result container for generated responses.
- Real-time display of backend results.

- Text area for user input
- Submit button
- CSS Styling (style.css):
 - Responsive design
 - Styled buttons, input areas, and result container

```
templates > index.html > ...
2 <html lang="en">
3 <head>
4 </head>
5 <body>
6 <h1>Welcome to EduGenie </h1>
7 <p>Your personal AI tutor for learning support!</p>
8
9 <!-- QnA -->
10 <form id="qaForm">
11 <label>Ask EduGenie a Question</label><br>
12 <input type="text" id="question" placeholder="Why is the sky blue?" required>
13 <button type="submit" value="Get Answer">Get Answer</button>
14 </form>
15 <div id="qaResult" class="output"></div>
16
17 <!-- Explaination -->
18 <form id="explainForm">
19 <label>Need an Explanation</label><br>
20 <input type="text" id="topic" placeholder="Photosynthesis" required>
21 <button type="submit" value="Explain">Explain</button>
22 </form>
23 <div id="explainResult" class="output"></div>
24
25 <!-- Summary -->
26 <form id="summaryForm">
27 <label>Summarize a Paragraph</label><br>
28 <input type="text" id="summaryText" placeholder="Paste long content to summarize" required>
29 <button type="submit" value="Summarize">Summarize</button>
30 </form>
31 <div id="summaryResult" class="output"></div>
32
33 <!-- Quiz -->
34 <form id="quizForm">
35 <label>Generate a Quiz</label><br>
36 <input type="text" id="quizText" placeholder="Enter content to generate quiz" required>
37 <button type="submit" value="Generate Quiz">Generate Quiz</button>
38 </form>
39 <div id="quizResult" class="output"></div>
40 </body>
41 </html>

```

Fig. Frontend integration from the project files

11. DEPLOYMENT

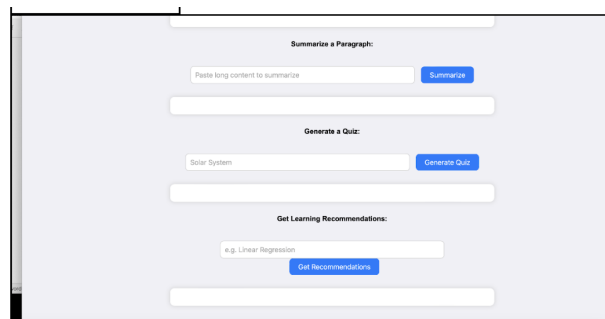
EduGenie can be run locally using Uvicorn. The documented command is: **uvicorn main:app --reload**. After starting the server, the application is accessed through the local address **http://127.0.0.1:8000**.

12. FUNCTIONAL TESTING

- Ask a question and verify the generated answer.
- Request an explanation of a topic.
- Generate a three-question MCQ quiz.
- Summarize educational content.
- Request a personalized learning plan.

13. APPLICATION USAGE

EduGenie supports several learning scenarios. A student can ask a factual question, request a quiz to check understanding of a topic, summarize a long passage for revision, or request a structured learning path for a subject such as SQL.



The screenshot shows the main interface of the EduGenie application. It features three distinct sections for user interaction:

- Summarize a Paragraph:** Includes a text input field labeled "Paste long content to summarize" and a blue "Summarize" button.
- Generate a Quiz:** Includes a text input field with "Solar System" as an example, and a blue "Generate Quiz" button.
- Get Learning Recommendations:** Includes a text input field with "e.g. Linear Regression" as an example, and a blue "Get Recommendations" button.

Fig. EDUGENIE

Exploring EduGenie:

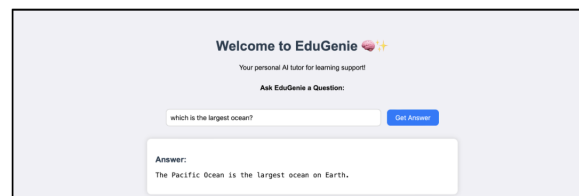
- Asking questions:

Fig. EduGenie main interface – actual output

Figure 13-1

Exploring EduGenie:

- Asking questions:



This screenshot displays the output of a question-answering interaction. At the top, it says "Welcome to EduGenie" with a heart and star icon, followed by "Your personal AI tutor for learning support!". Below this, the section "Ask EduGenie a Question:" contains a text input field with the question "which is the largest ocean?" and a blue "Get Answer" button. The answer is displayed in a box below: "Answer: The Pacific Ocean is the largest ocean on Earth."

Fig. Question-answering output

13.1 APPLICATION OUTPUT SCREENS

Explanation of a topic

Need an Explanation?

quantum computing

Explanation:

Quantum computing is a type of computing that uses the principles of quantum mechanics to perform certain calculations much faster than traditional computers. It allows computers to perform certain calculations much faster than traditional computers because it uses tiny particles called qubits that can exist in multiple states at the same time. This allows for new kinds of calculations and solving problems that are not possible with traditional computers.

Need an Explanation?

Binary Search Algorithm

Explanation:

The binary search algorithm is a way to find the index of a target value in a sorted list of numbers. It's like searching for a needle in a haystack. You need to find the number between 0 and 1, and then go back to the beginning of the list to find the target value.

Fig. Explanation output

Summarising long paragraphs

Summarize a Paragraph:

The Industrial Revolution, which began in the late 18th century, mark

Summary:

The Industrial Revolution changed the world by shifting it from farming to factories. New machines, like James Watt's steam engine, made things faster and easier to produce. While this boosted economies, it also created problems like bad working conditions, child labor, pollution, and overcrowded cities. Even with these problems, the Industrial Revolution shaped the modern world, affecting how we travel, communicate, work, and trade.

Fig. Summary output

Generating quizzes

It generates three questions with 4 options each. It corrects the answer if chosen wrong.

Generate a Quiz:

Pythagorean Theorem

Quiz:

Q1: What does the Pythagorean Theorem describe?

- ☐ The relationship between the angles of a triangle.
- ☐ The relationship between the sides of a right-angled triangle.
- ☐ The relationship between the area and perimeter of a triangle.
- ☐ The relationship between the sides of any triangle.

Q2: What does the Pythagorean Theorem describe?

The relationship between the angles of a triangle.

☒ The relationship between the sides of a right-angled triangle.

☐ The relationship between the area and perimeter of a triangle.

☐ The relationship between the sides of any triangle.

Correct

Q3: If a and b are the lengths of the two shorter sides of a right-angled triangle, and c is the length of the longest side (hypotenuse), what equation represents the Pythagorean Theorem?

$a + b = c$

☒ $a^2 + b^2 = c^2$

☐ $a^2 - b^2 = c^2$

☐ $2a + 2b = 2c$

Incorrect. Correct answer: $a^2 + b^2 = c^2$

Q4: Which type of triangle does the Pythagorean Theorem apply to?

Equilateral triangles

☒ Isosceles triangles

☐ Right-angled triangles

☐ All types of triangles

Correct

Fig. Quiz generation and answer checking

13.2 LEARNING RECOMMENDATIONS

The application guides learners with learning recommendations from beginner level to advanced level and provides details of resources and stepwise guidance.

e. Guiding with learning recommendations from beginners level to advanced level

Get Learning Recommendations:

SQL

Get Recommendations

Learning Recommendations for "SQL":

SQL Learning Path: From Zero to Hero

This learning path is structured to progressively introduce SQL concepts, starting from the basics and gradually advancing to more complex topics. It's designed to be adaptive: feel free to adjust the pace and delve deeper into areas that particularly interest you.

==I. Beginner Level: Building a Foundation==

==Estimated Time: 1-2 weeks==

• ==Key Topics==

- What is a Database and SQL? (Relational Model, DBMS)
- Basic Syntax (SELECT, FROM, WHERE)
- Data Types (INT, VARCHAR, DATE, etc.)
- Filtering Data (WHERE clause with comparison operators, logical operators: AND, OR, NOT)
- Ordering Results (ORDER BY)
- Limiting Results (LIMIT/OFFSET or TOP)
- Basic Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)

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- Ordering Results (ORDER BY)
- Limiting Results (LIMIT/OFFSET or TOP)
- Basic Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)
- Introduction to Tables and Columns

• ==Resources==

• ==Interactive Tutorials==

- SQLZoo: Excellent for hands-on practice with different SQL flavors.
- Khan Academy's SQL course: Offers a good introductory overview.
- Codecademy's Learn SQL: Interactive lessons with practical exercises.
- ==Videos==
- FreeCodeCamp's SQL Tutorial: Comprehensive and beginner-friendly.

==II. Intermediate Level: Working with Multiple Tables==

==Estimated Time: 2-3 weeks==

Fig. Learning recommendations output

EduGenie – Project Documentation

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14. ADVANTAGES

- Simple and accessible learning interface.
- Supports multiple educational tasks in one application.
- Provides concise explanations and summaries.
- Generates quizzes for self-assessment.
- Offers structured learning recommendations.
- Lightweight architecture suitable for limited hardware resources.

15. CHALLENGES

The project documentation identifies challenges including maintaining accuracy in AI-generated responses, performance across devices, personalization versus general usability, prompt tuning, UI clarity and data privacy.

16. FUTURE ENHANCEMENTS

- Voice-based interaction.
- Multilingual support.
- Mobile application with offline features.
- Progress tracking dashboards.
- Gamification with badges, streaks and challenges.
- Adaptive learning paths using data analytics.
- Group study sessions and teacher/parent dashboards.
- Integration with LMS platforms such as Moodle or Google Classroom.
- Image and PDF input recognition for doubt solving and summarization.
- Real-time synchronization and smart notifications.

17. CONCLUSION

EduGenie is an AI-powered educational assistant that combines generative AI with a simple web interface to support personalized learning. It simplifies complex concepts, provides question-and-answer assistance, generates quizzes, summarizes content and creates learning paths. The modular architecture also provides opportunities for future expansion such as voice interaction, multilingual learning, mobile support, progress tracking and adaptive learning.

REFERENCES

- Google AI for Developers – Gemini API documentation, as referenced in the project source.
- FastAPI documentation, as referenced in the project source.
- Jinja2 documentation, as referenced in the project source.
- PyPI/Uvicorn documentation, as referenced in the project source.